Statistical NLP: Lecture 4

Collocations

Thanks to Nathalie Japkowicz for slides

Introduction

- ✓ Collocations are characterized by limited ***compositionality***.
- ✓ Large overlap between the concepts of ***collocations*** and ***terms***, ***technical term*** and ***terminological phrase***.
- ✓ Collocations sometimes reflect interesting attitudes (in English) towards different types of substances: ***strong*** cigarettes, tea, coffee versus ***powerful*** drug (e.g., heroin)

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Definition (w.r.t Computational and Statistical Literature)

- [A collocation is defined as] a sequence of two or more consecutive words, that has characteristics of a syntactic and semantic unit, and whose exact and unambiguous meaning or connotation cannot be derived directly from the meaning or connotation of its components. [Chouekra, 1988]

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Other Definitions/Notions (w.r.t. Linguistic Literature) I

- ✓ Collocations are not necessarily adjacent
- ✓ Typical criteria for collocations: non-compositionality, non-substitutability, non-modifiability.
- ✓ Collocations cannot be translated into other languages.
- ✓ Generalization to weaker cases (strong association of words, but not necessarily fixed occurrence).

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Linguistic Subclasses of Collocations

- ✓ Light verbs: verbs with little semantic content
  - *make a decision* (not *take a decision)*
- ✓ Verb particle constructions or Phrasal Verbs
  - *to tell off, to go down*
- ✓ Proper Nouns/Names
- ✓ Terminological Expressions
  - *Hydraulic oil filter* (technical vocabulary)

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Overview of the Collocation Detecting Techniques Surveyed

- ✓ Selection of Collocations by ***Frequency***
- ✓ Selection of Collocation based on ***Mean and Variance*** of the distance between focal word and collocating word.
- ✓ ***Hypothesis Testing***
- ✓ ***Mutual Information***

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Frequency (Justeson & Katz, 1995)

1. Selecting the most frequently occurring bigrams
2. Passing the results through a part-of-speech filter

⇨ Simple method that works very well.

$C(w^1\ w^2)$	w^1	w^2
80871	of	the
58841	in	the
26430	to	the
21842	on	the
21839	for	the
18568	and	the
16121	that	the
15630	at	the
15494	to	be
13899	in	a
13689	of	a
13361	by	the
13183	with	the
12622	from	the
11428	New	York
10007	he	said
9775	as	a
9231	is	a
8753	has	been
8573	for	a

Table 5.1 Finding Collocations: Raw Frequency. $C(\cdot)$ is the frequency of something in the corpus.

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Part of Speech Tagging Filters

Tag Pattern	Example
A N	*linear function*
N N	*regression coefficients*
A A N	*Gaussian random variable*
A N N	*cumulative distribution function*
N A N	*mean squared error*
N N N	*class probability function*
N P N	*degrees of freedom*

Table 5.2 Part of speech tag patterns for collocation filtering. These patterns were used by Justeson and Katz to identify likely collocations among frequently occurring word sequences.

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$C(w^1\ w^2)$	w^1	w^2	tag pattern
11487	New	York	A N
7261	United	States	A N
5412	Los	Angeles	N N
3301	last	year	A N
3191	Saudi	Arabia	N N
2699	last	week	A N
2514	vice	president	A N
2378	Persian	Gulf	A N
2161	San	Francisco	N N
2106	President	Bush	N N
2001	Middle	East	A N
1942	Saddam	Hussein	N N
1867	Soviet	Union	A N
1850	White	House	A N
1633	United	Nations	A N
1337	York	City	N N
1328	oil	prices	N N
1210	next	year	A N
1074	chief	executive	A N
1073	real	estate	A N

Table 5.3 Finding Collocations: Justeson and Katz' part-of-speech filter.

Mean and Variance (Smadja et al., 1993)

- Frequency-based search works well for fixed phrases. However, many collocations consist of two words in more flexible relationships.
- The method computes the mean and variance of the offset (signed distance) between the two words in the corpus.
- If the offsets are randomly distributed (i.e., no collocation), then the variance/sample deviation will be high.

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Mean and Variance: Example

a. she knocked on his door

b. they knocked at the door

c. 100 women knocked on Donaldson's door

d. a man knocked on the metal front door

$$\frac{1}{4}(3+3+5+5) = 4.0$$

$$\sigma = \sqrt{\frac{1}{3}\left((3-4.0)^2 + (3-4.0)^2 + (5-4.0)^2 + (5-4.0)^2\right)} \approx 1.15$$

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σ	μ	Count	Word 1	Word 2
0.43	0.97	11657	New	York
0.48	1.83	24	previous	games
0.15	2.98	46	minus	points
0.49	3.87	131	hundreds	dollars
4.03	0.44	36	editorial	Atlanta
4.03	0.00	78	ring	New
3.96	0.19	119	point	hundredth
3.96	0.29	106	subscribers	by
1.07	1.45	80	strong	support
1.13	2.57	7	powerful	organizations
1.01	2.00	112	Richard	Nixon
1.05	0.00	10	Garrison	said

Table 5.5 Finding collocations based on mean and variance. Standard Deviation σ and mean μ of the distances between 12 word pairs.

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Hypothesis Testing I: Overview

- ✓ High frequency and low variance can be accidental. We want to determine whether the co-occurrence is random or whether it occurs more often than chance.
- ✓ This is a classical problem in Statistics called ***Hypothesis Testing***.
- ✓ We formulate a ***null hypothesis H_0*** (no association beyond chance) and calculate the probability that a collocation would occur if H_0 were true, and then ***reject H_0*** if p is too low and retain H_0 as possible, otherwise.

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Hypothesis Testing II: The *t* test

- The *t* test looks at the mean and variance of a sample of measurements, where the null hypothesis is that the sample is drawn from a distribution with mean μ.
- The test looks at the difference between the observed and expected means, scaled by the variance of the data, and tells us how likely one is to get a sample of that mean and variance assuming that the sample is drawn from a normal distribution with mean μ.
- To apply the *t* test to collocations, we think of the text corpus as a long sequence of N bigrams.

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The *t* test: An Example

$$P(new) = \frac{15828}{14307668}$$

$$P(companies) = \frac{4675}{14307668}$$

The null hypothesis is that occurrences of *new* and *companies* are independent.

$$\begin{aligned} H_0 : P(new\ companies) &= P(new)P(companies) \\ &= \frac{15828}{14307668} \times \frac{4675}{14307668} \approx 3.615 \times 10^{-7} \end{aligned}$$

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The *t* test: An Example (cont.)

- ✓ 8 occurrences of *new companies* in corpus

$$\bar{x} = \frac{8}{14307668} \approx 5.591 \times 10^{-7}$$

- ✓ Apply the *t* test

$$t = \frac{\bar{x} - \mu}{\sqrt{\frac{s^2}{N}}} \approx \frac{5.59110^{-7} - 3.61510^{-7}}{\sqrt{\frac{5.59110^{-7}}{14307668}}} \approx 0.999932$$

- ✓ 0.999932 < 2.576, 0.5% confidence level
  - ✓ *new companies* is not a collocation

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The *t* test works when frequency-based methods fail to distinguish

t	$C(w^1)$	$C(w^2)$	$C(w^1\ w^2)$	w^1	w^2
4.4721	42	20	20	Ayatollah	Ruhollah
4.4721	41	27	20	Bette	Midler
4.4720	30	117	20	Agatha	Christie
4.4720	77	59	20	videocassette	recorder
4.4720	24	320	20	unsalted	butter
2.3714	14907	9017	20	first	made
2.2446	13484	10570	20	over	many
1.3685	14734	13478	20	into	them
1.2176	14093	14776	20	like	people
0.8036	15019	15629	20	time	last

Table 5.6 Finding collocations: The t test applied to 10 bigrams that occur with frequency 20.

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Hypothesis Testing II: Hypothesis testing of differences (Church & Hanks, 1989

- We may also want to find words whose co-occurrence patterns best distinguish between two words. This application can be useful for Lexicography.
- The t test is extended to the comparison of the means of two normal populations.
- Here, the null hypothesis is that the average difference is 0.

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Hypothesis Testing of Differences

$$(5.4) \qquad t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}}}$$

t	$C(w)$	$C(strong\ w)$	$C(powerful\ w)$	word
3.1622	933	0	10	computers
2.8284	2337	0	8	computer
2.4494	289	0	6	symbol
2.4494	588	0	6	machines
7.0710	3685	50	0	support
6.3257	3616	58	7	enough
4.6904	986	22	0	safety

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Pearson's Chi-Square test I: Method

- ✓ Use of the t test has been criticized because it assumes that probabilities are approximately normally distributed (not true, generally).
- ✓ The Chi-Square test does not make this assumption.
- ✓ The essence of the test is to compare observed frequencies with frequencies expected for independence. If the difference between observed and expected frequencies is large, then we can reject the null hypothesis of independence.

Thanks to Nathalie Japkowicz for slides

Chi-Square Test: General Form

$$(5.6) \qquad X^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

- ✓ 2 by 2 table form

$$\chi^2 = \frac{N(O_{11}O_{22} - O_{12}O_{21})^2}{(O_{11} + O_{12})(O_{11} + O_{21})(O_{12} + O_{22})(O_{21} + O_{22})}$$

- ✓ Just like proving independence between two variables…

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Chi-Square Test: An Example

	$w_1 = new$	$w_1 \neq new$
$w_2 = companies$	8 (*new companies*)	4667 (e.g., *old companies*)
$w_2 \neq companies$	15820 (e.g., *new machines*)	14287181 (e.g., *old machines*)

$$\chi^2 = \frac{N(O_{11}O_{22} - O_{12}O_{21})^2}{(O_{11} + O_{12})(O_{11} + O_{21})(O_{12} + O_{22})(O_{21} + O_{22})}$$

This formula gives the following χ^2 value for Table 5.8:

$$\frac{14307668(8 \times 14287181 - 4667 \times 15820)^2}{(8 + 4667)(8 + 15820)(4667 + 14287181)(15820 + 14287181)} \approx 1.55$$

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Pearson's Chi-Square test II: Applications

- One of the early uses of the Chi square test in Statistical NLP was the identification of translation pairs in aligned corpora (Church & Gale, 1991).
- A more recent application is to use Chi square as a metric for corpus similarity (Kilgariff and Rose, 1998)
- Nevertheless, the Chi-Square test should not be used in small corpora.

Thanks to Nathalie Japkowicz for slides

Likelihood Ratios I: Within a single corpus (Dunning, 1993)

- ✓ Likelihood ratios are more appropriate for sparse data than the Chi-Square test. In addition, they are easier to interpret than the Chi-Square statistic.
- ✓ In applying the likelihood ratio test to collocation discovery, we examine the following two alternative hypotheses for the occurrence frequency of a bigram w1 w2:

- **Hypothesis 1.** $P(w^2|w^1) = p = P(w^2|\neg w^1)$
- **Hypothesis 2.** $P(w^2|w^1) = p_1 \neq p_2 = P(w^2|\neg w^1)$

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Likelihood Ratios for Collocations

	H_1	H_2
$P(w^2 \mid w^1)$	$p = \frac{c_2}{N}$	$p_1 = \frac{c_{12}}{c_1}$
$P(w^2 \mid \neg w^1)$	$p = \frac{c_2}{N}$	$p_2 = \frac{c_2 - c_{12}}{N - c_1}$
c_{12} out of c_1 bigrams are $w^1 w^2$	$b(c_{12};\ c_1, p)$	$b(c_{12};\ c_1, p_1)$
$c_2 - c_{12}$ out of $N - c_1$ bigrams are $\neg w^1 w^2$	$b(c_2 - c_{12};\ N - c_1, p)$	$b(c_2 - c_{12};\ N - c_1, p_2)$

$$
\begin{aligned}
\log \lambda &= \log \frac{L(H_1)}{L(H_2)} \\
&= \log \frac{b(c_{12}, c_1, p) b(c_2 - c_{12}, N - c_1, p)}{b(c_{12}, c_1, p_1) b(c_2 - c_{12}, N - c_1, p_2)} \\
&= \log L(c_{12}, c_1, p) + \log L(c_2 - c_{12}, N - c_1, p) \\
&\quad - \log L(c_{12}, c_1, p_1) - \log L(c_2 - c_{12}, N - c_1, p_2)
\end{aligned}
$$

where $L(k, n, x) = x^k (1 - x)^{n-k}$.

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Likelihood Ratios II: Between two or more corpora (Damerau, 1993)

- ✓ Ratios of ***relative frequencies*** between two or more different corpora can be used to discover collocations that are characteristic of a corpus when compared to other corpora.
- ✓ This approach is most useful for the discovery of subject-specific collocations.

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Mutual Information

- An Information-Theoretic measure for discovering collocations is ***pointwise mutual information*** (Church et al., 89, 91)
- Pointwise Mutual Information is roughly a measure of how much one word tells us about the other.
- Pointwise mutual information works particularly badly in sparse environments.

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Pointwise Mutual Information

$$(5.11) \quad I(x', y') = \log_2 \frac{P(x'y')}{P(x')P(y')}$$

$$(5.12) \quad = \log_2 \frac{P(x'|y')}{P(x')}$$

$$(5.13) \quad = \log_2 \frac{P(y'|x')}{P(y')}$$

- ✓ Other things being equal, bigrams composed of low-frequency words will receive a higher score than bigrams composed of high-frequency words

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$I(w^1, w^2)$	$C(w^1)$	$C(w^2)$	$C(w^1\ w^2)$	w^1	w^2
18.38	42	20	20	Ayatollah	Ruhollah
17.98	41	27	20	Bette	Midler
16.31	30	117	20	Agatha	Christie
15.94	77	59	20	videocassette	recorder
15.19	24	320	20	unsalted	butter
1.09	14907	9017	20	first	made
1.01	13484	10570	20	over	many
0.53	14734	13478	20	into	them
0.46	14093	14776	20	like	people
0.29	15019	15629	20	time	last

Table 5.14 Finding collocations: Ten bigrams that occur with frequency 20, ranked according to mutual information.

$$I(Ayatollah, Ruhollah) = \log_2 \frac{\frac{20}{14307668}}{\frac{42}{14307668} \times \frac{20}{14307668}} \approx 18.38$$

Thanks to Nathalie Japkowicz for slides